



TEST 2 FSPM0034 SEM 1 2122

Statistics and Probability (Universiti Teknologi Malaysia)



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Foundation Programme
Universiti Teknologi Malaysia
Session 2021/2022 Semester 1

Test 2 (20%)

Time: 1 hour 45 minutes

FSPM0034

Answer all questions. Show all your work clearly.

1. Let X be a random variable with the following probability distribution as follows;

x	0	1	2	3	4
$P(X = x)$	0.2	a	0.5	0.1	b

- a) If $E(X) = 1.95$, find the values of a and b . (4 marks)
- b) Find the probability of X is not less than 2. (2 marks)
- c) Determine $E(2X + 5)$. (2 marks)
2. A continuous random variable X has the cumulative distribution function given by;

$$F(x) = \begin{cases} 0, & x \leq 1 \\ \frac{(x-a)^2}{10}, & 1 \leq x \leq 3 \\ \frac{12x - x^2 - 21}{b}, & 3 \leq x \leq 6 \\ 1, & x \geq 6 \end{cases}$$

where a and b are constants.

- a) Find the values of a and b . (3 marks)
- b) Find $P(2 \leq X < 4)$. (2 marks)
- c) Determine the probability density function of X , $f(x)$. (3 marks)
- d) Sketch the graph of $y = f(x)$. (2 marks)

3. Given the probability density function of X is;

$$f(x) = \begin{cases} \frac{1}{12}(x + 1); & 0 \leq x \leq 4 \\ 0; & \text{otherwise} \end{cases}$$

Find:

- a) $E(X)$ (2 marks)
 - b) $Var(X)$ (3 marks)
 - c) $Var(3X - 5)$ (2 marks)
4. A carton contains 24 boxes where each box contains 15 identical computer parts. 90% of these computer parts are non-defective. Find the probability that;
- a) a box has all non-defective computer part. (2 marks)
 - b) less than two boxes in a carton have all non-defective computer parts. (3 marks)
 - c) A shipment of these computer parts are sent to a vendor. The vendor run a test to determine whether to accept or reject the shipment by testing the computer parts in 100 cartons selected at random. If more than 6 cartons contain less than 2 boxes that have no defective computer parts, then the shipment is rejected. Find the probability that the shipment is rejected. (3 marks)
5. A group of students from the Medical Faculty carries out some experiments during their field trip. They took a 10-liter bucket of water from a swamp. They found an average of 80 mosquito larvae in the water.
- a) Write down the probability mass function with its parameter for the above question. Justify why you choose that kind of distribution. (2 marks)
 - b) Compute the probability that between eight and fifteen mosquito larvae contain in a 1-liter bucket of water. (3 marks)
 - c) A 300 ml flask of water is taken from the bucket for further analysis. Find the probability that the flask contains at least four mosquito larvae? (2 marks)

End

LIST OF FORMULA FSPM 0034

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\sigma^2 = \frac{1}{N} \left(\sum_{i=1}^N x_i^2 - \frac{\left(\sum_{i=1}^N x_i \right)^2}{N} \right)$$

$$\sigma = \sqrt{\sigma^2}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$s^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} \right)$$

$$s = \sqrt{s^2}$$

$$\mu = \frac{\sum_{i=1}^N f_i x_i}{\sum_{i=1}^N f_i}$$

$$\sigma^2 = \frac{1}{\sum_{i=1}^N f_i} \left(\sum_{i=1}^N f_i x_i^2 - \frac{\left(\sum_{i=1}^N f_i x_i \right)^2}{\sum_{i=1}^N f_i} \right)$$

$$\sigma = \sqrt{\sigma^2}$$

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

$$s^2 = \frac{1}{\sum_{i=1}^n f_i - 1} \left(\sum_{i=1}^n f_i x_i^2 - \frac{\left(\sum_{i=1}^n f_i x_i \right)^2}{\sum_{i=1}^n f_i} \right)$$

$$s = \sqrt{s^2}$$

$$\text{Median : } L + \left(\frac{\frac{n}{2} - F}{f} \right) \times c$$

$$\text{Mode: } L + \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right) \times c$$

$$P_i = L + \left(\frac{\frac{i}{100}(n) - F}{f} \right) \times c$$

$$Q_i = L + \left(\frac{\frac{i}{4}(n) - F}{f} \right) \times c$$

$$D_i = L + \left(\frac{\frac{i}{10}(n) - F}{f} \right) \times c$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) > 0.$$

$${}^n C_r = \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

$${}^n P_r = \binom{n}{r} = \frac{n!}{(n-r)!}$$

Random Variable

$$E(X) = \sum xP(X = x)$$

$$E(X) = \int xf(x)dx$$

$$Var(X) = E(X^2) - [E(X)]^2$$

Probability distributions

Distribution	Probability functions	Mean	Variance
Binomial	$P(X = x) = \binom{n}{x} \pi^x (1 - \pi)^{n-x}$ $x = 0, 1, \dots, n, \quad 0 \leq \pi \leq 1$	$n\pi$	$n\pi(1 - \pi)$
Poisson	$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ $x = 0, 1, 2, \dots, \quad \lambda > 0$	λ	λ
Normal	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$ $-\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \sigma > 0.$	μ	σ^2
Standard Normal	$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$		